1. Verify that $[p \to (q \to r)] \to [(p \to q) \to (p \to r)]$ is tautology

→ The following truth table proves required result

p	q	r	$p \to q$	$q \to r$	$(p \to q) \wedge (q \to r)$	$p \to r$	$(p \to q) \wedge (q \to r) \to (p \to r)$
0	0	0	1	1	1	1	1
0	0	1	1	1	1	1	1
0	1	0	1	0	0	1	1
0	1	1	1	1	1	1	1
1	0	0	0	1	0	0	1
1	0	1	0	1	0	1	1
1	1	0	1	0	0	0	1
1	1	1	1	1	1	1	1

2. Using laws of logic prove the logical equivalence

$$p \vee q \vee (\neg p \wedge \neg q \wedge r) \iff p \vee q \vee r$$

⇒ We have

$$p \vee q \vee (\sim p \wedge \sim q \wedge r)$$

⇒ $\sim p \wedge \sim q \wedge r \iff (\sim p \wedge \sim q) \wedge r$, by Associative law

$\iff \sim(p \vee q) \wedge r$, by DeMorgan Law.

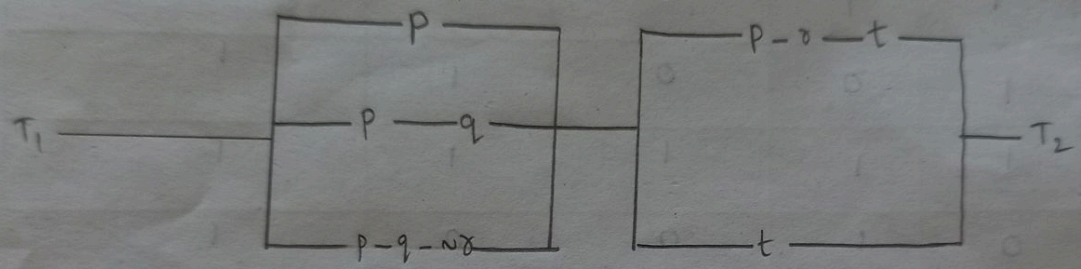
$\therefore\ p \vee q \vee (\sim p \wedge \sim q \wedge r) \iff (p \vee q) \vee [\sim (p \vee q) \vee r]$

$\iff [(p \vee q) \vee \sim (p \vee q)] \wedge [(p \vee q) \vee r]$, by Distribution Law

$\iff T_0 \wedge (p \vee q \vee r)$, by Inverse Law and Associative Law

$\iff p \vee q \vee r$, by Commutative and Identity Laws.

3. Simplify the network shown in the figure



> By examining the given network, we find that it is represented by $u \equiv v \wedge w$ where

$$v \equiv \{p \vee (p \wedge q) \vee (p \wedge q \wedge \rightarrow r)\}$$

$$w \equiv (p \wedge r \wedge t) \vee t$$

We note that

$v \iff p \vee \{p \wedge (q \wedge \rightarrow r)\}$ by using absorption and associative laws

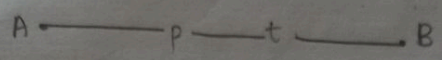
$\iff p$, by absorption law

and $w \iff t \vee \{t \wedge (p \wedge r)\}$, by using commutative and associative laws

$\iff t$, by absorption law

$\therefore u \equiv v \wedge w \iff p \wedge t$

This shows that the given network which has 10 switches is equivalent to a network with only two switches p and t in a series. The simplified network is shown below.



4. Verify the following argument

$$\begin{array}{l} (\sim p \vee q) \rightarrow r \\ r \rightarrow (s \vee t) \\ \sim s \rightarrow \sim u \\ \sim u \rightarrow \sim t \\ \hline \therefore p \end{array}$$

i) We find that

$[(\sim p \vee q) \rightarrow r] \wedge [r \rightarrow (s \vee t)] \wedge [\sim s \wedge \sim u) \wedge (\sim u \rightarrow \sim t)$

$\Rightarrow [(\sim p \vee q) \rightarrow (s \vee t)] \wedge \sim s \wedge \{\sim u \wedge (\sim u \rightarrow \sim t)\}$

by Rule of Syllogism and associative law

$\Rightarrow [(\sim p \vee q) \rightarrow (s \vee t)] \wedge [\sim s \wedge \sim t]$, by Modus Ponens Rule

$\Leftrightarrow [(\sim p \vee q) \rightarrow (s \wedge t)] \wedge [\sim (s \vee t)]$, using DeMorgan Law

$\Rightarrow \sim (\sim p \vee q)$, by Modus Tollens Rule

$\Leftrightarrow p \wedge \sim q$

$\Rightarrow p$, by rule of Conjuctive simplification

04

5. Let $p(x)$ be an open statement "$x^2 = 2x$" where the universe comprises of all integers. Determine whether each of the following statements is true or false.

a) $p(0)$

b) $p(1)$

c) $p(2)$

d) $p(-2)$

e) $\exists x\, p(x)$

f) $\forall x\, p(x)$

a) $p(0): 0^2 = 2 \cdot (0) \Rightarrow p(0): 0 = 0$ This statement is obviously true.

b) $p(1): 1^2 = 2 \cdot 1 \Rightarrow p(1): 1 = 2$ This statement is obviously false $(1 \neq 2)$

c) $p(2): 2^2 = 2 \cdot 2 \Rightarrow p(2): 4 = 4$ This statement is obviously true.

d) $p(-2): (-2)^2 = 2 : (-2) \Rightarrow p(-2): 4 = -4$ This statement is obviously again false.

e) $\exists x\, p(x)$: There exists an integer x for which $x^2 = 2x$. Since 2 is an integer, and $2^2 = 2 \cdot 2$ this statement is true.

f) $\forall x\, p(x)$: For all integers x, $x^2 = 2x$. Since 1 is an integer a $1^2 \neq 2 \cdot 1$, this statement is false (the negation for this statem is $\exists x\, \sim p(x)$, which we have showed is true by this exam -therefore, the statement is false.

6. Prove the validity of the argument given

$$\frac{\begin{array}{l}\forall x\ [p(x) \vee q(x)] \\ \forall x\ [\sim p(x) \wedge q(x) \rightarrow r(x)]\end{array}}{\therefore\ \sim r(x) \rightarrow p(x)}$$

⇒ Suppose $\sim r(x)$ is true for any x in the universe. Then we find that

$$[\{\sim p(x) \wedge q(x)\} \rightarrow r(x)]$$

$\Leftrightarrow \sim r(x) \rightarrow \{p(x) \vee \sim q(x)\}$ using Contrapositive and DeMorgan Law

$\Rightarrow p(x) \vee \sim q(x)$, by Modus Ponens Rule

Therefore for any x,

$$[p(x) \vee q(x)] \wedge [\{\sim p(x) \wedge q(x)\} \rightarrow r(x)]$$

$\Rightarrow [p(x) \vee q(x)] \wedge [p(x) \vee \sim q(x)]$

$\Leftrightarrow p(x) \vee [q(x) \wedge \sim q(x)]$, by distributive law

$\Leftrightarrow p(x)$, because $q(x) \wedge \sim q(x)$ is false

Thus when $\sim r(x)$ is true, the given premises imply $p(x)$. Therefore $\sim r(x) \rightarrow p(x)$ is a true statement. In view of the Rule of Universal Generalization, this proves that given argument is valid.

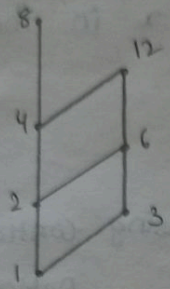
06

7. For $A = \{1,2,3,4,6,8,12\}$ the Hasse diagram for POSET (A,R) is given in the figure

a) Determine the relation matrix

b) Construct the directed graph G on A

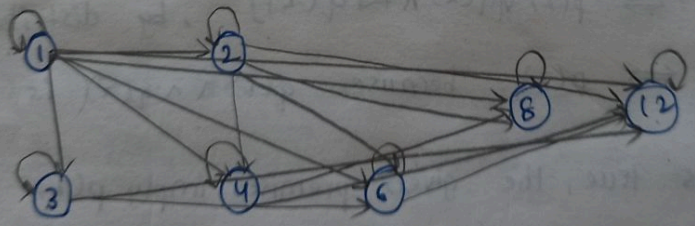
c) Topologically sort the POSET (A,R)



⇒ By examining the given Hasse diagram, we note that

a) $R = \{(1,1),(2,2),(3,3),(4,4),(6,6),(8,8),(12,12),(1,3),(1,2),(1,6),(1,4)$
$(1,12),(1,8),(3,6),(3,12),(2,6),(2,4),(2,12),(2,8),(6,12),$
$(4,8),(4,12)\}$

b)



a) Relation Matrix

$M(R) =$

	1	2	3	4	6	8	12
1	1	1	1	1	1	1	1
2	0	1	0	1	1	1	1
3	0	0	1	0	1	0	1
4	0	0	0	1	0	1	1
6	0	0	0	0	1	0	1
8	0	0	0	0	0	1	1
12	0	0	0	0	0	0	1

c) By examining the Hasse diagram of R, we find that vertices that are to be identified as $v_1, v_2, v_3, v_4, v_5, v_6, v_7$ & v_8 in the topological sorting process are

8. For $A = \{1,2,3,4\}$, let R and S be the relations on A defined by $R = \{(1,2), (1,3), (2,4), (4,4)\}$ and $S = \{(1,1), (1,2), (2,3), (2,4)\}$, find $R \circ S$, $S \circ R$, R^2, R^3, S^2, S^3.

$\Rightarrow$ $R \circ S = \{(1,3), (1,4)\}$

$S \circ R = \{(1,2), (1,3), (1,4), (2,4)\}$

$R^2 = R \circ R = \{(1,4), (2,4)\}$

$R^3 = R^2 \circ R = \{(1,4), (2,4)\}$

$S^2 = S \circ S = \{(1,2), (1,1), (1,3), (1,4)\}$

$S^3 = S^2 \circ S = \{(1,3), (1,4), (1,1), (1,2)\}$

08

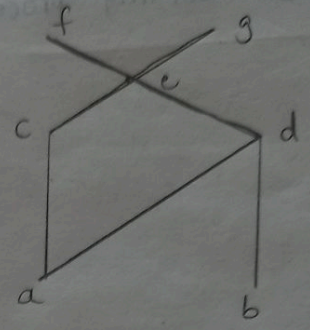
9. Consider the Hasse diagram of a POSET (A, R) given in figure. If $B = \{c, d, e\}$ find the following

a) Upper bound

b) Lower bound

c) Least upper bound

d) Greatest lower bound.



a) Upper bound = e, f, g

b) Lower bound = a

By examining all upward paths from c, d, e in the given Hasse diagram, we find that LUB (B) = e. By examining all upward paths to c, d, e we find that GLB (B) = a.

10. Define the encoding function $E : \mathbb{Z}_2^3 \rightarrow \mathbb{Z}_2^6$ by means of the parity check matrix

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

a) Determine all the code words

b) Does this code correct all single errors in transmission?

a) We have

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

which is of the form $[A^T | I_3]$. Accordingly

$$A^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{so that} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Hence the associated generator matrix is

$$G = [I_3 | A] = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$[E(000)] = [0\ 0\ 0]\, G = 0$$

$$[E(001)] = [0\ 0\ 1] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= [0\ 0\ 1\ 1\ 0\ 1]$$

$$[E(010)] = [0\ 1\ 0] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= [0\ 1\ 0\ 0\ 1\ 0]$$

$$[E(011)] = [0\ 1\ 1] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} = [0\ 1\ 1\ 1\ 1\ 1]$$

10

$$[E(100)] = [1\ 0\ 0]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= [1\ 0\ 0\ 1\ 1\ 1]$$

$$[E(101)] = [1\ 0\ 1]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= [1\ 0\ 1\ 0\ 0\ 0]$$

$$[E(110)] = [1\ 1\ 0]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= [1\ 1\ 0\ 1\ 0\ 0]$$

$$[E(111)] = [1\ 1\ 1]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= [1\ 1\ 1\ 0\ 0\ 0]$$

b) We observe that two columns of H (namely the 2nd and 5th) are identical. Therefore H does not provide a decoding scheme that corrects single errors in transmission.

11. In the figure given below determine

i) A walk from n to p that is not a trail

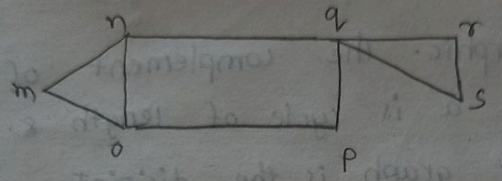
ii) A n-p trial that is not a path

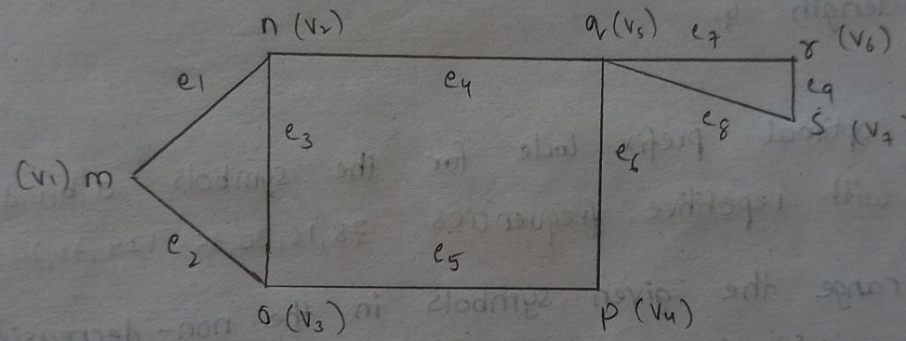
iii) path from n to p

iv) A closed walk from n to n that is not circuit

v) A circuit from n to n that is not cycle

vi) A cycle from n to n





i) $ne_4qe_7re_9se_8qe_4ne_3oe_5p$

ii) $ne_3oe_2me_1ne_4qe_6p$

iii) ne_3oe_5p

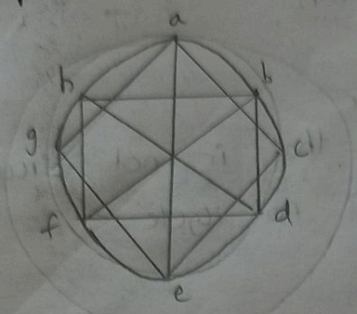
iv) $ne_4qe_7re_9se_8qe_4n$

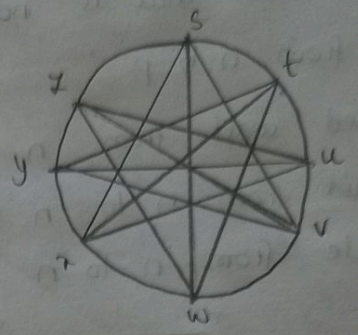
v) $ne_3oe_5pe_6qe_7re_9se_8qe_4$

vi) $ne_1me_2oe_3n$

12

12. Determine whether the graphs given are isomorphic or not? If isomorphic write their mappings.





⇒ They are not isomorphic. The complement of the graph containing vertex a is cycle of length 8. The complement of the other graph is the disjoint union of two cycles of length 4.

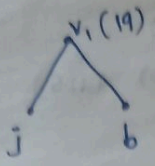
13- Construct an optimal prefix code for the symbols a,b,c,d,e,f,g,h,i,j that occurs with repetitive frequencies 78,16,30,35,125,31,20,50,80,3

⇒ First we arrange the given symbols in the non-decreasing order of their weights (frequencies). Their representation as isolated vertices is shown below:

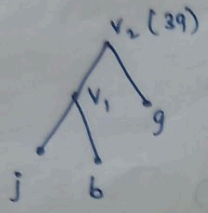
j	b	g	c	f	d	h	a	i	e
(3)	(16)	(20)	(30)	(31)	(35)	(50)	(78)	(80)	(125)

We now construct an optimal tree having a,b,c,d,e,f,g,h,i,j as leaves by using the Huffman procedure. The construction is done step-by-step and the graphs obtained in these

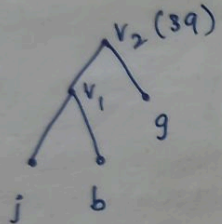
steps are shown below in the order of their occurence. The final graph is labeled.

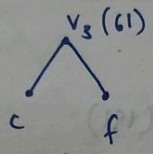


g c f d h a i e
(20) (20) (31) (35) (50) (78) (80) (125)

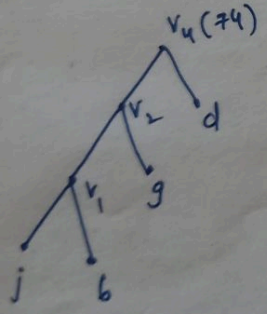


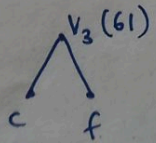
c f d h a i e
(20) (31) (35) (50) (78) (80) (125)



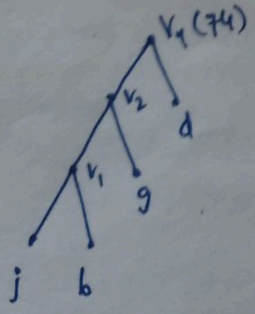


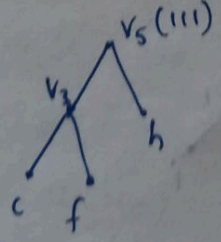
d h a i e
(35) (50) (78) (80) (125)





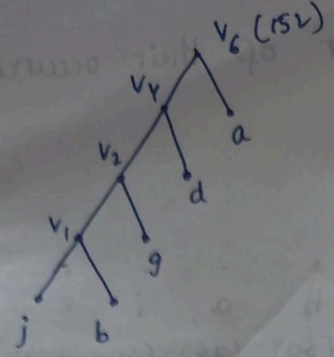
h a i e
(50) (78) (80) (125)

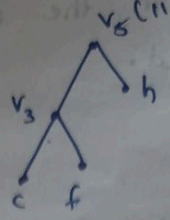




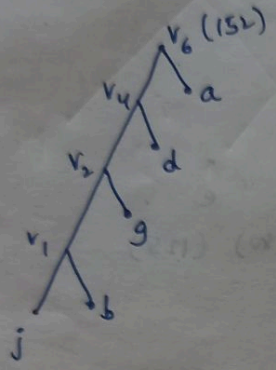
a I e
(78) (80) (125)

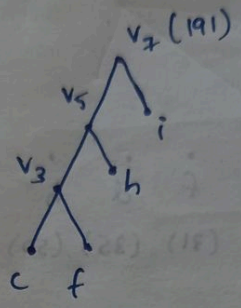
14



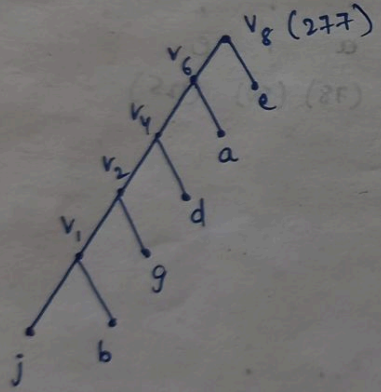


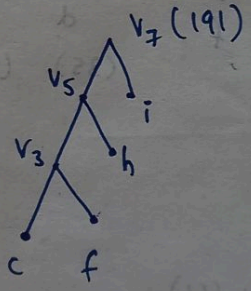
i (80) e (125)

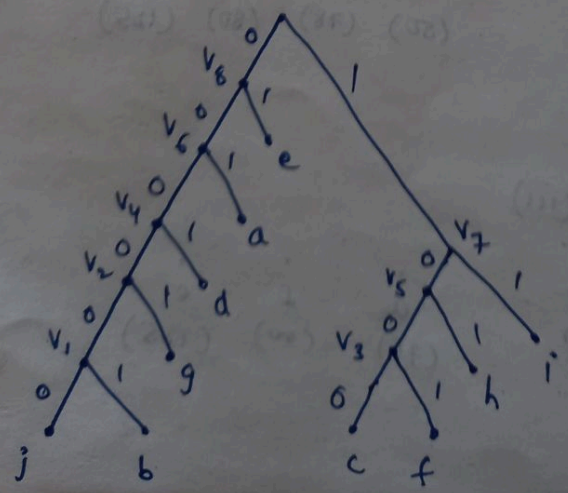




e (125)







The tree in figure is an optimal tree for the given data. From this tree, we obtain the following optimal prefix code for the given symbols.

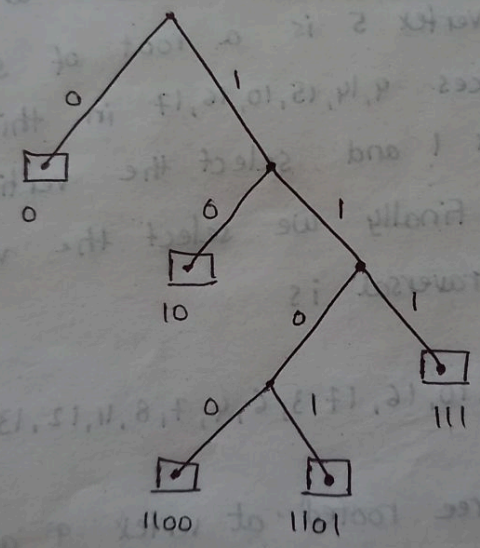
A : 001	B : 000001	C : 1000	D : 0001	E : 01
F : 1001	G : 00001	H : 101	I : 11	J : 000000

14. For the prefix code given in the figure decode the sequences

a) 100111101

b) 10111100111000111101

c) 11011111100010

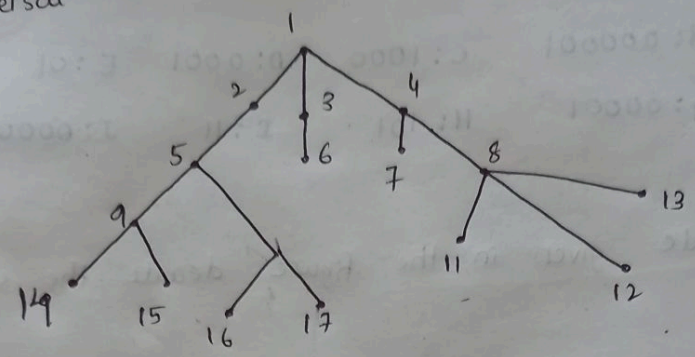


i) Splitting : 10 0 111 1101 , Decode : e b a d

ii) Splitting : 10 111 10 0 1100 0 1101 , Decode : e a e b c b

iii) Splitting : 1101 111 1100 10 Decode : d a c e

15. List the vertices in the tree shown in figure when they are visited in a preorder traversal and in a postorder traversal.



⇒ First we select the root of tree namely 1. Next we select the vertex 2. Since there is no subtree rooted at 2, we go to the vertex 5. The vertex 5 is a root of subtree. In this subtree, we select vertices 9, 14, 15, 10, 16, 17 in this order. Then we return to the vertex 1 and select the vertices 3 and 6 and return to vertex 1. Finally we select the vertices 4, 7, 8, 11, 12, 13. Thus the preorder traversal is

1, 2, 5, 9, 14, 15, 10, 16, 17, 3, 6, 4, 7, 8, 11, 12, 13

We start with the subtree rooted at vertex 9 and select the vertices 14, 15 and 9 in this order. Then, we go to the subtree rooted at 5 and select the vertices 16, 17, 10 and 5 in this order. Then we select 2. Since 2 is not a root, we go to the subtree that lies to the right. Here, 13, 8, 4. Finally we select root of the tree. Thus postorder traversal is

14, 15, 9, 16, 17, 10, 5, 2, 6, 3, 7, 11, 12, 13, 8, 4, 1